hw1.py > ...

```python
# 1a)

def fib(n):
    if (n < 2):   # O(1)
        return 1    # O(1)
    else:
        return fib(n-1) + fib(n-2)*fib(n-3) # O(2^n)

print(fib(5))

# The time complexity of this function is O(2^n) due to the exponential growth of recursive calls.


# 1b)

# fib2(n)
    # initialize the base cases of 0, 1, and 1 = 1
    # create counters for addintions and multiplications
    # for loop from 3-n:
        # calcuate the next number in the seqeuence
        # increase the counters for additions and multiplications
    # return the nth number in the sequence along with the counters
```

written

0.2) show that, if $c > 0$, then $g(n) = 1 + c + c^2 + c^n$ is

a) $\Theta(1)$ if $c < 1$ $\quad g < g(n) < x \quad 1 > 1 - c^{n+1} > 1 - c$

$1 + \sum_{i=1}^{n} c^i$ $\quad \left[\frac{1}{1-c} > g(n)_{c<1} > 1 \right.$

if $c := 1$

$g(n) = \frac{1 - c^{n+1}}{1 - c} = \frac{c^{n+1} - 1}{c - 1}$

b) $\Theta(n)$ if $c = 1$ $\quad 1 \le c^n \le n$

$1 \le c^n \le n$ which is $\Theta(n)$ since everything is 1

c) $\Theta(c^n)$ if $c > 1$

$c - 1 < c^{n+1} - 1 < c^{n+1}$

$\left[1 < \frac{c^{n+1} - 1}{c - 1} < \frac{c^{n+1}}{c - 1} \right.$

which is $\Theta(c^n)$ since $c-1$ doesn't effect c^n